

Modern Stylometry in the Age of Generative AI: Interpretability, Formal Stuckness, and Cultural Collapse

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Abstract

The proliferation of AI-generated text presents challenges that current stylometric methods are ill-equipped to address. Stylometry has long excelled at author discrimination but has prioritised accuracy at the expense of interpretability. This review argues that this gap, long tolerated in human authorship attribution, becomes an urgent practical problem in the age of AI-generated prose, and that recent advances in structural and syntactic feature methods offer a path toward addressing it. The review situates this gap within the broader societal consequences of AI prose proliferation, including adversarial misuse, cultural homogenisation, and the systemic degradation of linguistic diversity. Building on Heuser’s concept of “formal stuckness” in AI-generated verse, it extends the framework to AI prose, and argues that “cultural collapse”, the systematic flattening of formal diversity across generative systems, is a pervasive property of AI-generated text. Three research questions are proposed, designed to characterise the AI prose stylome in interpretable terms, alongside the methodological approach required to address them.

1 Introduction

Generative AI has, in the space of a few years, become perhaps the most prolific author in human history. That this text is largely indistinguishable from human writing is not merely a technical curiosity; it is a problem with concrete cultural, political, and epistemic consequences. Addressing these consequences requires not just powerful tools for identifying AI-generated text, but a formal account of what makes it different from human writing.

The field best positioned to provide such an account is stylometry: the computational analysis of authorial style through measurable formal features. Stylometry as a field has existed since the late 19th century, and has developed powerful tools for author discrimination, demonstrating that authorial

style leaves measurable traces in text, even across large and varied corpora. This builds on what has become known as the “fingerprint” tradition - the idea that authors leave unconscious, measurable traces in their writing that persist across texts and can be reliably identified. Yet this discriminative power has come at a cost: the field has optimised relentlessly for accuracy, but has made little progress in understanding what these measurable differences actually consist of in linguistic terms. The fingerprint tradition can measure the fingerprint, but cannot say what the fingerprint means (Stamatatos, 2009).

This review argues that, with the rise of generative AI, stylometry is at a critical juncture. The interpretability gap that has long been present in the field becomes an urgent problem when applied to AI-generated text, as it prevents the field from articulating what formally distinguishes AI text from human writing. Recent work by Heuser (2025a) has begun to address this gap in the context of AI-generated verse, identifying patterns of “formal stuckness” - a tendency toward syntactic and prosodic conservatism - and “cultural collapse” - the systematic flattening of formal diversity across generative systems. Heuser’s findings provide the theoretical and empirical foundation for the research questions proposed here, and this review extends his framework to the domain of AI prose, which remains formally uncharacterised.

Section 2 traces the historical development of stylometric methods. Sections 3 and 4 examine structural and syntactic approaches that begin to address the interpretability gap. Section 5 applies these frameworks to AI-generated text via Heuser’s work. Section 6 documents the broader societal consequences of AI prose proliferation. Section 7 proposes three research questions designed to characterise the AI prose stylome in interpretable terms, and outlines the methodological approach required to address them.

2 The Fingerprint Tradition

The discipline of stylometry and the broader field of computational literary analysis trace their roots back to the 19th century, as an attempt to apply scientific methods to answer questions brought up through the study of literature. Some of the earliest work in this area was done by [Mendenhall \(1887\)](#), when he demonstrated that statistical distributions of word length could distinguish between authors, establishing the foundational premise that style leaves a quantifiable and reproducible trace in text. This approach, and the new field of stylometry it helped to establish, progressed through the early-to-mid 20th century, with significant contributions from scholars such as Yule and Zipf identifying mathematical regularities in prose as potential authorial markers ([Stamatatos, 2009](#)).

[Mosteller and Wallace \(1984\)](#) applied these principles to the question of the authorship of the Federalist Papers, using a Bayesian approach to attribute disputed essays to either Hamilton or Madison based on word frequency patterns. Mosteller and Wallace first attempted to use sentence length to solve this problem, but found that it was not a reliable discriminator, with both the average sentence length and the standard deviation of sentence lengths being very similar across the two authors ([Mosteller and Wallace, 1984](#), §1.3). To solve this problem, they turned to the frequency rates of function words, which they saw as good candidates for authorial markers because they are less likely to be influenced by context (although they commented that defining what is and is not a contextual word is a difficult problem itself, often requiring a degree of subjective judgement) ([Mosteller and Wallace, 1984](#), §2.1). They found that the frequency rates of function words were a much more reliable discriminator, and by applying a Bayesian model to these frequencies, they were able to attribute the disputed essays with a high degree of confidence, attributing all 12 disputed essays to Madison ([Mosteller and Wallace, 1984](#), §9.1). Yet Mosteller and Wallace themselves acknowledged that no theory existed relating these frequency patterns to an author's personality or writing style - the method excelled at identification, but offered no account of what those patterns actually meant in terms of how an author writes ([Mosteller and Wallace, 1984](#), §9.3).

Building on this approach, [Burrows \(2002\)](#) introduced the "Delta" measure, which quantifies the distance between an unknown text and a set of

known texts by computing the mean absolute deviation between z-scored word frequency profiles. This method demonstrated remarkable attribution accuracy across a large corpus of Restoration poetry and has become a standard tool. Crucially, however, Burrows openly acknowledged that the method's success rests on principles that remain poorly understood - producing a reliable number, but not an explanation. What unites these approaches is their focus on identifying a unique stylome or "fingerprint" for each author, based on the assumption that an author's style is consistent and can be captured through quantifiable features. This concept is highlighted in the work of [Van Halteren et al. \(2005\)](#), who extended these techniques beyond canonical literary corpora to a random sample of student authors. The study proved this approach worked, achieving 97.8% attribution accuracy on a deliberately challenging corpus. Notably, Van Halteren found that syntactic features alone achieved 88.7% accuracy in their experiments - far higher than might be expected from features that resist straightforward linguistic interpretation. However, despite this predictive power, neither the stylome concept nor the methods used to demonstrate it offer any account of what these measurable differences actually consist of in linguistic terms.

In more recent decades, the field has evolved to incorporate a wide range of stylometric features, which [Stamatatos \(2009\)](#) organises into a taxonomy ranging from lexical to syntactic. The lexical approaches tokenise text into words and analyse their frequencies, with the most common words (function words) often proving to be the most informative for attribution tasks (because they are used unconsciously and are less likely to be influenced by topic ([Koppel and Schler, 2003](#))). Character-level features, specifically character n-grams, capture sequences of characters and can thus retain sub-word information, punctuation patterns, and morphological habits that word-level approaches miss. Syntactic features, including part-of-speech sequences, parse tree rewrite rules and dependency patterns, promise a richer account of authorial style, but have consistently underperformed lexical approaches in practice, largely due to noise introduced by automated parsing tools.

What this taxonomy reveals is a field that has optimised relentlessly for discrimination at the expense of explanation. Character n-grams achieve impressive accuracy precisely because they are opaque - no literary scholar can read a character

n-gram frequency profile and say anything meaningful about how an author writes. As [Stamatatos \(2009\)](#) acknowledges, the field remains unable to provide a higher-level description of authorial style: it can measure the fingerprint, but cannot say what the fingerprint means.

3 Refinement and Internal Critique

Despite the state-of-the-art methods (such as Burrows' Delta) achieving outstanding accuracy in author discrimination tasks, there is remarkably little understanding of how it works, and more importantly, *why it works*. [Argamon \(2008\)](#) was the first to provide a geometric explanation, showing that Delta is mathematically equivalent to Manhattan distance applied to z-scored word frequencies, and connecting it to a probabilistic interpretation based on the Laplace distribution. Building on this foundation, [Evert et al. \(2017\)](#) tested Argamon's framework systematically, revealing and emphasising this gap between the ubiquity of the method and its understanding - specifically pointing out "There is almost no other algorithm in stylometry that has been used as much as Delta and still there is no theoretical framework to explain its success". Evert et al. found that the mathematically "correct" fixes suggested by Argamon's framework - using Laplace distribution parameters or Euclidean distance - performed no better than Burrows's original formulation, deepening rather than resolving the mystery. The hypothesis that the pattern of over- and under-utilisation of vocabulary was more important than the amplitude of the z-scores was confirmed upon testing, allowing one to understand what Burrows' Delta is sensitive to, but still offers no account of why those patterns are authorial. The fact that Delta is measuring a pattern of hundreds of tiny word frequency deviations presents a fundamental problem for interpretability, as it makes it very hard to point to a single feature to explain why a text was attributed to a certain author. Even more importantly, it means that it becomes impossible to describe the difference between one author's profile and another in any humanly meaningful terms - a reader could not connect Delta's output to anything observable in the text.

The fingerprint tradition has now interrogated itself mathematically and found that even a geometric understanding of Delta does not translate into linguistic explanation. The authorial signal is confirmed to exist, Van Halteren proved that, but

what that signal consists of in terms of actual writing choices remains opaque. If frequency profiles across hundreds of words are uninterpretable by definition, then interpretability has to come from a different kind of feature altogether, one that operates at the level of linguistic structure rather than statistical distance.

4 Towards Structural Features

Collecting and analysing structural features of prose has been a long-standing goal in stylometry, as they promise a more interpretable account of authorial style. This has been present in the field over the last few decades, with [Baayen et al. \(1996\)](#) being some of the earliest to attempt to use syntactic features for attribution. They extracted syntactic rewrite rule frequencies from fully parsed texts and demonstrated that these outperformed lexical measures, whilst acknowledging the requirement for a sophisticated and accurate parser. [Gamon \(2004\)](#) extended this approach using deep linguistic analysis features including syntactic and semantic information, finding that whilst syntactic features alone underperformed lexical approaches, their combination yielded improved attribution accuracy. Both approaches, however, remained firmly within the discriminative tradition - syntax was treated as another feature set for improving classification accuracy, with no attempt to characterise what those syntactic patterns actually revealed about an author's writing.

Probabilistic Context-Free Grammars (PCFGs) model sentence structure as a hierarchical parse tree of syntactic constituents: noun phrases, verb phrases, subordinate clauses, capturing not just what words appear in a text but how sentences are architecturally constructed. Each production rule in the grammar (e.g. $S \rightarrow NP VP$) is assigned a probability, and the probability of a complete parse tree is computed as the product of the probabilities of the rules used in its derivation, making individual rules extractable as discrete, countable features ([Jurafsky and Martin, 2009](#), §14.1, §14.1.1). However, as PCFGs treat each rule application as statistically independent of its context in the tree, extracted rule frequencies do not capture the structural dependencies between constituents that a linguistically motivated analysis would recognise ([Jurafsky and Martin, 2009](#), §14.4).

However, whilst PCFGs have seen some use in stylometry, they have not yet been able to de-

liver on their promise of interpretability. This gap is explored in [Feng et al. \(2012\)](#), who observe that whilst prior work using PCFG models had demonstrated that syntactic structure carries authorial signal, the individual production rules extracted by such models resist meaningful interpretation. A rule such as $S^{\text{ROOT}} \rightarrow \text{SBAR}, \text{NP VP}$ indicates something about clause ordering, but does not map onto any concept that a rhetorician or literary scholar would recognise as a stylistic category. [Feng et al. \(2012\)](#) addresses this by treating the parse tree not as a feature space in itself, but as a substrate from which humanly legible features can be derived. They operationalise two well-established rhetorical distinctions: the simple/complex/compound/complex-compound taxonomy and the loose/periodic contrast, as algorithmic procedures over PCFG trees, producing features that a literary scholar could recognise and engage with meaningfully rather than opaque statistical profiles.

However, whilst [Feng et al. \(2012\)](#) demonstrates that interpretable syntactic features are both achievable and authoritatively distinctive in human writing, the question of whether such features can characterise the difference between human and AI-generated prose remains entirely unaddressed - a gap that becomes increasingly urgent as large language models produce text at scale.

5 AI Prose as a New Test Case

The rise of Large Language Models (LLMs) has added new emphasis on this interpretability gap. Whilst with human authorship attribution, the practical stakes of opacity are relatively low - a Delta score that correctly attributes a poem to Milton without explaining why is still useful - with AI-generated text, opacity becomes a genuine social problem. An informed public discourse around AI requires not just the ability to identify AI-generated text, but to articulate why it feels so different from human writing.

Contemporary AI detection systems demonstrate that the opacity problem has not been resolved but merely reproduced at higher accuracy. [Emi and Spero \(2024\)](#) report 99% classification accuracy across ten text domains using a transformer-based neural network, yet the system produces a probability score rather than an explanation. It can identify AI-generated text, but without saying whether that identification rests on syntactic regularity, lexical

flatness, prosodic repetition, or some combination thereof. The central failing of the fingerprint tradition has been completely and fully inherited by the AI detection tradition - the field has a powerful tool for discrimination, but no account of what it is discriminating.

As discussed earlier, the fingerprint tradition demonstrated that human authors have their own unique stylome, characterising how they write. LLMs appear to have developed their own stylome, one that both critics and everyday readers have developed an intuitive recognition of. A colloquial term has emerged to describe this stylome: "slop". This recognition, and its colloquial name, has become widespread in public discourse, and AI text is often identified by specific tics such as the eager-to-please register and syntactic repetition. However, just like human stylomes did before Burrows, this intuition has resisted formal characterisation. "Slop" has often been seen rather derisively in academic circles, and as a vague cultural criticism. [Heuser \(2025b\)](#) is one of the first scholars to truly treat it as a legitimate object of inquiry, asking what measurable properties of AI verse produce this recognition. The question of interpretability becomes slightly different with AI: we need to characterise not an individual author's idiosyncrasies, but a generative system's systematic tendencies.

In an initial attempt to explore LLM generated text, [Heuser \(2025b\)](#) explored patterns of AI generated poetry versus historical, human written, poetry, in an attempt to explain why these generated poems often appear so "sloppy", and hollow. When analysing historical poetry, it was found that rhyme in the English verse dropped from around 80% to just 7% in post-war poetry. And yet, when prompting AI for poetry without specifying rhyming, AI produced rhyming poetry 95% of the time (there was in fact little difference between when AI was specifically asked for rhyming in the prompt, and when rhyming was not mentioned). This figure exceeds even the most formally conservative periods of literary history. Perhaps even more strikingly, when specifically prompted not to rhyme, averaged across the tested models, LLMs still produced rhyming verse around 50% of the time. Heuser terms this "formal stuckness" - generative verse appears to prefer older, more formal conventions, implementing them even when told not to, and at a higher rate than was ever shown throughout history. Heuser posits that this comes from LLMs abstracting smooth generalised patterns from histor-

ical variation. Crucially, the naming and defining of this "formal stuckness" shows that Heuser points to a specific, humanly legible feature (rhyme rate), and can use it to show where AI text diverges from human writing, and how much. Much as with the recent attempts at interpretable human stylometry, the analyses on AI text are becoming interpretable. But arguably, the analyses discussed here show that it is perhaps going beyond the basic interpretability approaches that are becoming common in human stylometry, and is providing something that the fingerprint tradition has thus far failed to do: producing a specific finding a literary critic could engage with directly, and showing why that finding matters to readers. The finding of "formal stuckness" is not just a number, but a specific claim about what AI text does and does not do, and why that matters to readers. This contrasts with the fingerprint tradition - a reader or critic cannot look at a delta score and say anything meaningful about what it means for a text to have a delta of 0.5 versus 0.2, but they can look at a rhyme rate of 95% and say "this is a very high rhyme rate, even higher than the most formally conservative periods of history, and this is something that makes AI verse feel different from human verse". This is the kind of interpretability currently lacking in stylometry - not just powerful tools for discrimination, but specific, humanly legible claims about what makes an author's style unique, or what makes AI text feel different from human text.

Heuser then goes deeper in his analyses when assessing meter across AI poetry, specifically Shakespeare style sonnets. Shakespeare wrote lines with unambiguous iambic pentameter just 24% of the time, whereas LLM imitations produced this unambiguous iambic pentameter 53% of the time, an improvement in metrical regularity by more than a factor of 2, and once again exceeding even the most formally conservative historical periods (Heuser, 2025b). These models have learnt the formal structure of Shakespearean sonnets, whilst dramatically underrepresenting the conventional exceptions that characterise the genre. This is perhaps even more strongly shown in what the models omit: trochaic inversions. These appear in 12.8% of Shakespeare's lines, but only 1.9% of LLM's. When visualised across a metrical space, LLM-written sonnets occupy a more concentrated metrical space than not just an average of human authors, but more concentrated than just Shakespeare alone. This finding is important as it reveals a new layer

on which the AI stylome is based, not simply the presence of features, but the systematic absence of variation. Furthermore, it shows the work and the findings that truly interpretable stylometry can accomplish: not simply some black-box probability score, but a specific claim, nameable and describable, showing what AI text lacks and why that absence matters to readers.

Having gone some way in establishing formally what the AI stylome looks like, the question becomes why is it this way, *why* is LLM writing so "sloppy". The most intuitive answer would be the biases of its training data: LLMs overuse rhyme and produce metrically hyper-regular verse simply because their training data over-represents traditional, formally conservative verse. Heuser directly tests this hypothesis, and finds that it does not hold (Heuser, 2025b). His analysis of poems found in LLM training data found that they are not disproportionately rhyming, and in some cases found the poems rhymed in the training data rhymed less than average. AI training data generally focuses more heavily on contemporary poetry, and yet LLM writing seems more reflective of traditional, historical poetry. This finding points towards something more fundamental in how these models process and generate forms of art. When comparing two different versions of the open source model Llama 3.1, its instruction-tuned variant, and its base model, Heuser found the instruction-tuned model rhymes approximately 60% of the time on postwar poem completions, compared to 10% for the base model. Instruction tuning is a process designed to make models output content that humans will both find more aesthetically pleasing, and more helpful. And yet, this process appears to push models towards a stricter reliance on older, more formal structures and choices that the model perceives as more conventional and pleasing. The pure fact that the AI stylome is not simply a technical artifact, but is shaped by economic and social pressures to produce content humans find pleasing is significant - it suggests that what readers perceive as "slop" is not an unintended glitch but a structurally produced outcome, whose causes are as much commercial as computational, and whose formal conservatism is, in a very real sense, by design (though the precise mechanisms underlying this tendency remain an open question). In his companion paper (Heuser, 2025a), Heuser extends this argument into the concept of "cultural collapse". He defines this as a computational logic that prioritises satis-

faction over frustration, regularity over variation, and conformity over contradiction. This is found to be pervasive not only in aesthetic form, but also more general historical representation - when asked to generate lists of authors published at Random House between 1950 and 2000, models produced demographic distributions showing 50% white authorship, against a documented historical reality of 90-95%. The AI stylome is not an accident of training data but a systematic product of how these systems are built, pointing to something more fundamental about what generative systems can and can't represent.

Fredric Jameson says that for a poem to have meaning beyond simply legibility, it must be a "socially symbolic act" (Jameson 1981, cited in Heuser 2025b), meaning that it should be more than simply a hallucination of syntax and grammar, and should be built on a background of social and historical contradictions. Where intentionalist critics have argued that AI-generated texts lack meaning because they lack an author, Heuser uses Jameson's definition of poetry to propose that generative texts lack meaning not because they lack an author, but because they lack a *history*. Once all a model's training data is dissolved into its neural network, generative text no longer has the set of social and historical contradictions to mediate. As Heuser puts it, outputs that are simultaneously nostalgic and ahistorical reflect neither a past nor present cultural reality and distort true historical context in an attempt to produce "the average of our aesthetic desires".

Jameson's definition raises a parallel concern: if AI text lacks history, then frequency based detection methods that rely on patterns of word usage may be structurally insufficient for characterising the AI stylome, as they are not designed to capture the absence of historical context. Instead, features that capture the *absence* of historical variation are needed, and this is exactly what syntactic and prosodic features can provide.

Heuser proposes "generative formalism" as a tool to examine how LLMs process and flatten cultural production. This critical framework offers a theoretical grounding for interpretable stylometric analysis of generative text. Crucially, however, Heuser's findings are entirely focused on poetic verse, whether these patterns of a reduction in syntactic variation and structural conservatism operate on the broader level of AI prose remains entirely unexamined. The formal properties of the AI prose

stylome, unlike those of AI verse, have not yet been named. The tools discussed earlier in sections 3 and 4 of this review: structural features and interpretive syntactic motifs, have not yet been applied to AI prose, and are potentially the tools needed to characterise the AI prose stylome in a way that is both interpretable and meaningful to readers.

6 The Stakes of Interpretability

Whilst Heuser's findings are formally constrained to verse, the phenomena he describes, of a 'formal stuckness' and 'cultural collapse', operate across every written register (Heuser, 2025a). Within professional writing alone, a survey by Britain's Society of Authors found that 20% of fiction and 25% of nonfiction writers were already using generative AI for some of their work (Kriss, 2025), and this figure captures only the most visible stratum of a transformation that extends across journalism, corporate communication, academic writing, and social media. The significance of this transformation, however, lies not merely in its extent but in the nature of what is being proliferated. Bender et al. (2021) argue that human language takes place between individuals who share common ground and have communicative intents which they use language to convey. Human language relies on shared context, models of the world, and assumptions about the reader's mental state to convey implicit meaning. However, they argue that LLM text is produced without any communicative intent and without grounding in a model of the world or the reader's mental state. LLM text is form without substance, a statistical recombination of linguistic patterns that produces fluency without meaning. Human perception of language is mediated by a predisposition to interpret communicative acts as conveying meaning and intent, and if one side of that communication has no meaning, then the reader's comprehension of the implicit meaning is merely an illusion. At the scale described above, the formal properties of AI prose carry consequences that extend well beyond the aesthetic, and yet they remain formally uncharacterised.

The most urgent and concrete danger of the ever-increasing ubiquity of AI prose is adversarial use. As Goldstein et al. (2023) discuss, the ability to generate large volumes of text that is indistinguishable from human writing has significant implications for misinformation, disinformation, and propaganda. For bad actors looking to spread propa-

ganda, LLMs offer a tool to automate the production of large volumes of convincing, yet misleading text. As the technology progresses, for these bad actors, LLMs will only become more usable, reliable and efficient. But these dangers are not confined to organised political influence operations. [Bender et al. \(2021\)](#) identify LLMs as a potential tool to aid in extremist recruitment and social media manipulation, and language models reproducing abusive content makes it more likely that targets of abuse will be exposed to more of it, and face serious psychological harm. A further danger lies in machine translation reproducing output that is inaccurate, and yet fluent to a reader who may not have access to or understand the source text, apparent in the case of a Palestinian man who was arrested by Israeli authorities after a Facebook post of his that said "good morning" in Arabic was translated as "attack them" in Hebrew ([Bender et al., 2021](#)). These harms are not incidental misuses of the technology, but direct consequences of the structural property identified above - fluency without communicative grounding, received by readers predisposed to trust it.

As AI-generated text floods the internet and becomes incorporated into training data for future models, [Shumailov et al. \(2024\)](#) demonstrate that this causes irreversible defects in the resulting models. Crucially, indiscriminate use of model-generated content in training causes a significant, and potentially irreversible drop in performance across a range of tasks, and causes the tails of the original content distributions to disappear, leading to a loss of diversity and a flattening of the resulting models' outputs - [Heuser's \(2025a\)](#) concept of "cultural collapse". What Heuser observed in verse - the gravitational pull toward formal conservatism and the suppression of variation - is a systemic property of generative systems. As AI-generated text becomes more prevalent and continues to contaminate future training data, the rare and diverse voices at the edge of the distribution are the first to be eliminated. Cultural collapse is not merely an aesthetic observation about current AI output - it describes the trajectory of written culture as a whole. But, beyond formal diversity, this argument also applies to a loss of ideological diversity. [Bender et al. \(2021\)](#) note that LLMs absorb the hegemonic worldview of their training data, and as model-generated text increasingly constitutes that training data, those biases are not merely preserved but amplified through successive generations of

models. The feedback loop operates on not only formal, but also ideological dimensions, narrowing both the range of styles and the range of perspectives available in written culture. The consequences of this are not merely abstract and systemic, they are already legible in the texture of everyday writing.

This process is visible even at the level of a single punctuation mark—the em-dash offers a case study in cultural collapse operating at the level of a single formal feature. Even the most casual of readers have, by now, noticed that AI-generated text overuses em-dashes to an almost hilarious degree, inserting them whether or not it suits the context it is writing in, and resisting user's attempts to get LLMs to stop using them. This has become so widely prevalent that the em-dash is now the most widely recognised marker of AI-generated text, with users seeing one and immediately assuming AI authorship, and a common trope in public discourse about the "sloppiness" of AI prose ([Kriss, 2025](#)). The register incongruity of this becomes clear when considering the practical constraints on human em-dash usage. For a writer composing a social media post on a standard smartphone or laptop keyboard, an em-dash is not natively accessible; producing one requires deliberately searching for and copying the character, an effort almost no writer would make for an informal post. The em-dash has historically been confined to formal and academic writing precisely because these are the contexts in which writers have both the tools and the stylistic motivation to deploy it. In $\text{\LaTeX}$, the formatter widely used for scientific and mathematical writing, an em-dash is produced simply by typing three consecutive hyphens, making it perhaps the most natural punctuation choice available in that environment. AI, by contrast, deploys it indiscriminately across all registers, a direct instance of the formal conservatism [Heuser \(2025a\)](#) describes, in which statistical correlation with valued prose overrides sensitivity to context. The consequences of this are already felt by human writers. The present author, writing this review in $\text{\LaTeX}$ where em-dashes are trivially easy to produce, found a persistent reluctance to use them throughout - not because they are grammatically inappropriate in academic prose, but because an academic reader encountering one has no prior reason to trust the authenticity of the writing, and may instinctively read it as a marker of AI authorship. A feature with a legitimate centuries-long history in

formal writing has been effectively contaminated. This is cultural collapse at the level of a single formal feature: AI’s formal conservatism has not merely homogenised its own output, but has begun to constrain the stylistic choices available to human writers. This is perhaps the most concrete illustration of why formally characterising the AI stylome matters. That the present author’s attempt to use a language model to redraft this very paragraph resulted in the immediate reintroduction of em-dashes is, in this context, instructive rather than merely ironic.

The consequences documented in this section, and the highlighted dangers of AI-generated text, show why detection alone is fundamentally insufficient. The ability to identify AI-generated text is important, but knowing that a text is AI-generated does not reveal what makes it that way, and without a formal account of what the AI stylome actually consists of, it is impossible to address any of the consequences described above at their root. In adversarial contexts, black-box detection classifiers are brittle, and easily circumvented by small changes to the text. It is impossible to understand or resist what cannot be formally characterised. If we want to understand how to meaningfully address the harms of AI prose proliferation, we need to understand what makes it different from human writing in the first place. This requires a formal characterisation of the AI stylome, a humanly legible description of the formal properties of AI prose. The research questions outlined in the next section represent the first steps towards that goal.

7 Open Questions and Proposed Research Directions

The dangers and consequences of AI-generated prose documented in the preceding section make the interpretability gap in stylometry not merely an academic concern, but an urgent practical one. Across the field of stylometry, there is a clear and persistent gap between the ability to discriminate between authors (or between human and AI text) and the ability to explain what those differences actually consist of in linguistic terms. The operationalisation of structural features as interpretable stylometric measures, demonstrated by Feng et al. offers a promising avenue for developing interpretable stylometric analyses, and Heuser has shown this can be extended to generated text, in the context of AI-generated poetry (Heuser, 2025b).

However, the formal properties of AI prose remain unexamined, and there is a pressing need to apply interpretable stylometric methods to this domain to understand the systematic tendencies of AI-generated prose and their implications for readers and for the broader cultural and political consequences documented above.

RQ1: Does AI prose show reduced variance and greater regularity across interpretable syntactic features, such as loose/periodic sentences and clause complexity, when compared to human prose?

Feng et al.’s method for operationalising the loose/periodic sentence distinction and the simple, complex, compound, complex-compound taxonomy has been demonstrated to be both statistically distinctive and humanly legible in human writing. However, it remains an open question whether these features can characterise the difference between human and AI prose. Heuser has shown that AI verse is more metrically regular and less prosodically varied than human verse, but the question of whether similar patterns of regularity and reduced variance are present in AI prose remains unexamined. Applying Feng et al.’s interpretable syntactic features to a corpus of AI-generated prose would therefore establish whether AI prose exhibits these tendencies towards regularity and reduced variance. If AI prose does exhibit reduced variance and greater regularity across these features, this would provide a specific, humanly legible account of the AI stylome in prose, and would help to explain why AI prose feels different from human prose to readers. It would also suggest that the concept of "cultural collapse" identified by Heuser in AI verse extends to prose, and that interpretable stylometric analysis can provide the formal vocabulary to describe what readers already intuitively recognise as the AI stylome. Establishing this formal vocabulary is not merely descriptive; it is a prerequisite for addressing the cultural, adversarial, and epistemic harms documented in the preceding section. Of the three questions proposed here, RQ1 represents the most immediate research priority, as the syntactic features it examines form the analytical foundation upon which RQ2 and RQ3 build.

RQ2: Does LLM-generated prose show a regularisation in rhythm (such as sentence length variance, cadence and stress patterns) compared to human prose, analogous to what Heuser found in verse?

Heuser’s analysis of AI verse revealed a signifi-

cant regularisation in both rhyme and meter. However, it remains an open question whether similar patterns of regularisation are present in AI prose. Applying prosodic measures to AI-generated prose (sentence length variance, cadence, and stress patterns) would establish whether LLMs also exhibit a tendency towards rhythmic regularity. These are features a reader can perceive directly: the "flatness" of AI prose that people notice intuitively may be measurable at the rhythmic level. Unlike verse meter, prose rhythm lacks a formal template, making operationalisation more challenging, but also potentially more revealing. If such regularisation is found, it would suggest that the intuitive perception of 'flatness' in AI prose has a measurable formal basis - that what readers sense but cannot articulate may be quantifiable at the rhythmic level, and that interpretable prosodic analysis can surface it. The intuitive perception of AI prose as flat or lifeless is precisely the phenomenon whose cultural and epistemic consequences have been documented above; a formal account of its rhythmic basis would therefore carry significance well beyond stylometric methodology.

RQ3: Does the process of instruction tuning amplify formal conservatism when applied to AI-generated prose, similarly to how it affects formal features of verse such as rhyme and meter?

Heuser's finding that instruction tuning amplifies formal conservatism in AI verse raises the question of whether this effect also applies to AI prose. Instruction tuning is designed to make models produce content that is more aesthetically pleasing to humans, but given its effect on verse, it may also push models towards more conventional and regular patterns in prose. Comparing instruction-tuned and base versions of LLMs on measures of syntactic regularity and rhythmic variation in prose would establish whether instruction tuning has a similar effect on prose as it does on verse. If instruction tuning is found to amplify formal conservatism in prose, this would suggest that the commercial and social pressures to produce pleasing content are driving models towards even greater regularity and reduced variation, further contributing to the phenomenon of "cultural collapse" in generative systems, compounding the adversarial, cultural, and epistemic consequences documented in the preceding section.

All three of these research questions share a common requirement: a cross-sectional corpus design (Rasinger, 2013) in which human and AI prose are

sampled at a single point in time across multiple domains and models. The multi-domain requirement is particularly important, because as shown by Baayen et al. (1996), register variation can mask or override the signal of authorial style; if all the human prose in the corpus is from a single domain (e.g. news articles) and all the AI prose is from a different domain (e.g. fiction), then any differences found may be due to register rather than the human/AI distinction. Moreover, research question 3 specifically requires access to both instruction-tuned and base versions of LLMs, following Heuser's methodology but extended to prose. This is a novel corpus construction that does not yet fully exist at scale. Such a corpus would require paired human and AI-generated prose sampled across multiple domains and model types, including both instruction-tuned and base model variants to address RQ3; a full specification of candidate datasets is provided in Appendix A. The application of these features to AI-generated prose constitutes not merely the natural next step for interpretable stylometry, but a necessary response to the cultural, adversarial, and epistemic consequences documented in the preceding section.

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A Candidate Datasets

This appendix lists the candidate datasets for the empirical work proposed in Section 7.

Table 1: Candidate datasets for the proposed research questions.

Dataset	Type	Domain	Size	RQs	Access	Notes
ACL Anthology Reference Corpus	Human	Academic prose	~10k papers	RQ1, RQ2	aclanthology.org	Used by Feng et al. (2012) ; published baselines available
Project Gutenberg	Human	Literary prose	>60k texts	RQ1, RQ2	gutenberg.org	Public domain novels; used by Feng et al. (2012) ; published baselines available; researchers typically use a small subset
Ghostbuster	Mixed (human + AI)	News, student essays, creative writing	~21k docs	RQ1, RQ2	github.com/vivek3141/ghostbuster-data	1k human + 5k ChatGPT + 1k Claude per domain (3 domains); human text from IvyPanda, Reuters 50-50, and r/WritingPrompts; multi-domain design mitigates register confound (Baayen et al., 1996); only dataset here with a direct public download link (Verma et al., 2024)
Llama 3.1 8B (base)	AI-generated (base model)	Prose (to be constructed)	TBD	RQ3	HuggingFace: meta-llama/Llama-3.1-8B	Self-constructed corpus; prompts adapted from Heuser (2025b) for prose domain
Llama 3.1 8B Instruct	AI-generated (instruct model)	Prose (to be constructed)	TBD	RQ3	HuggingFace: meta-llama/Llama-3.1-8B-Instruct	Paired with base model above for direct instruction tuning comparison